

Optimising Power Systems in Scotland

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1 Motivation

For over 100 years, the UK has heavily relied on coal power stations to generate electricity. But this era has come to an end, on the 30th of September 2024 the UK closed its last coal power station [1]. A truly historic moment in the fight against global warming. As we look into the future, in order to meet climate change demands the UK will have to shift to more renewable power generation. To achieve this, the UK could rely more on hydroelectric power. A team from the Australian National University (RE100) created an atlas identifying suitable locations for hydroelectric plants [2]. By using this atlas to select potential "Bluefield" sites¹, and filtering for stations with a capacity of 15 GWh and an 18-hour maximum capacity duration we generate Figure 1:

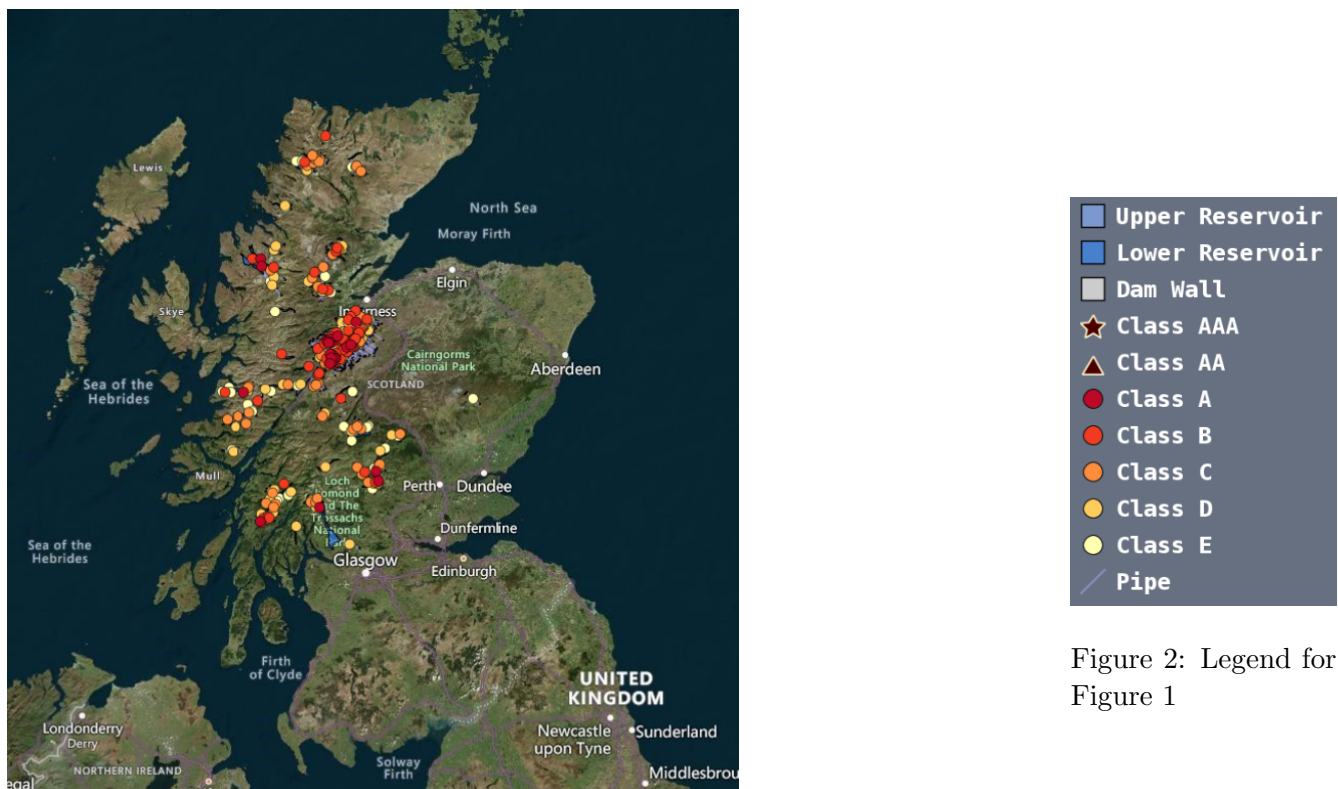


Figure 2: Legend for Figure 1

Figure 1: Map of Scotland with indicators at suitable locations for hydroelectric power plants [2].

Making use of Figure 2 we can see the potential locations that would be most efficient in both cost and production. All of the generated power plants are on unprotected land, this minimises potential damage caused by implementing new power plants. Now we have seen the potential of hydroelectric plants in Scotland, the next step would be working to optimise the operation of such plants.

¹Bluefield sites refers to all potential hydroelectric power stations that use at least one existing reservoir.

2 Optimisation of systems

2.1 Formulating the problem

The selection of Bluefield sites means that the new systems will be formed of multiple interconnected reservoirs which flow into each other. At any point we have to make a decision about how much water we will release from one reservoir, which could have potential consequences. If we choose to release too much of the water supply we risk it not being replenished in time to meet future demand, meaning we have to rely on other more expensive generation methods. On the other hand if we decide to limit the flow of water we risk reservoirs overflowing, not only damaging the environment but also increasing running costs of the plant.

We will first cover some terminology around dynamic programming and how this relates to our problem. We divide the whole problem into **stages**, where a decision has to be made at each stage. In our case, each stage will be a set of times where we want to restrict, increase or maintain water flow. Given our weather prediction is only accurate in relatively short periods of time our stages will be as follows, $t = \{1, \dots, T\}$ where T can be adjusted based off your desired accuracy [3].

For each stage there is a set of **states**, these describe possible conditions of the corresponding stage. In context of hydroelectric systems, they will be composed of the current capacity of the reservoir and the history of inflow from previous stages $\mathbf{x}_t = (\mathbf{v}_t, \mathbf{a}_{t-1}, \mathbf{a}_{t-2}, \dots, \mathbf{a}_1)$, where $\mathbf{v}_t$ is the reservoir storage levels and $\mathbf{a}_t$ is the inflow volume of water at time t [3].

We require a **decision variable** we can change in order to control the value of the objective function (defined below). For us, this will be comprised of the outflow volume and the spillage volume. So it will be of the form $\mathbf{u}_t = (\mathbf{q}_t, \mathbf{s}_t)$ where $\mathbf{q}_t$ is the outflow volume and $\mathbf{s}_t$ is the spilled volume, both at time t .

To update information between stages we use a **transition equation**. For reservoir systems, this is equivalent to updating the capacity of reservoirs, since any water that flows from one reservoir to another will change their state at the next stage. So we will need the current capacity, inflow, outflow and spilled volumes. For any given station, we want to know which reservoirs are contributing to the inflow and which are receiving the outflow. Using an incidence matrix $\mathbf{M}$, we can create a transition equation that works for all reservoirs in any system.

$$\mathbf{v}_{t+1} = \mathbf{v}_t + \mathbf{a}_t + \mathbf{M}(\mathbf{q}_t + \mathbf{s}_t).$$

Finally, we need to define constraints on our system. Ensuring the water level is at most full capacity we have, $\mathbf{v}_{t+1} \leq \mathbf{v}_{max}$. Practically, the outflow will be restricted by the turbines so $\mathbf{q}_t \leq \mathbf{q}_{max}$. We can include a condition for the lower bound of the outflow because if there is any spillage we know that, $\mathbf{q}_t + \mathbf{s}_t \geq \mathbf{q}_{min}$ [3].

Now we have all the components we need to create a stochastic dynamic programming (SDP) problem. Our objective is to minimise costs, however one could easily alter this to maximise power. We construct the objective function as the expectation of the minimum cost for the current stage plus the objective function at the next stage [3]. Writing our complete problem out we have,

$$\alpha_t(\mathbf{x}_t) = \mathbb{E}_{\mathbf{a}_t | \mathbf{x}_t} \left\{ \min_{\mathbf{u}_t} c_t(\mathbf{u}_t) + \beta \alpha_{t+1}(\mathbf{x}_{t+1}) \right\} \quad (1)$$

subject to

$$\mathbf{v}_{t+1} = \mathbf{v}_t + \mathbf{a}_t + \mathbf{M}(\mathbf{q}_t + \mathbf{s}_t), \quad (2)$$

$$\mathbf{v}_{t+1} \leq \mathbf{v}_{max}, \quad (3)$$

$$\mathbf{q}_t \leq \mathbf{q}_{max}, \quad (4)$$

$$\mathbf{q}_t + \mathbf{s}_t \geq \mathbf{q}_{min}. \quad (5)$$

$\forall t = T, T-1, \dots, 1, \forall \mathbf{x}_t$, where:

t : indexes the stages (T end time),
 $\mathbf{x}_t$: state vector at the beginning of stage t ,
 $\alpha_t(\mathbf{x}_t)$: expected value of operation cost from state $\mathbf{x}_t$,
 $\mathbf{a}_t$: inflow volume,
 $\mathbf{q}_t$: outflow volume,
 $\mathbf{s}_t$: spilled volumes,
 $\mathbf{M}$ incidence matrix,
 $\mathbf{a}_t|\mathbf{x}_t$: probability distribution of $\mathbf{a}_t$, given state $\mathbf{x}_t$
 $\mathbb{E}\{\cdot\}$: expected value,
 $\mathbf{u}_t$: decision vector for stage t ,
 $\mathbf{c}_t(\cdot)$: immediate cost due to decision $\mathbf{u}_t$,
 β : discount factor,
(2): state transition equation,
(3): constraint on water capacity,
(4), (5): constraints on the outflow,

Notice how the objective function (1) is recursively defined, we can use dynamic programming techniques to solve this. The general is as follows, first we solve the last case for all possible states, then we use that information to solve the second to last case. We iterate this until we have an optimal solution starting from the first case.

2.2 Solving the deterministic problem

Since we have constructed the hydroelectric application into a SDP problem, we can now begin to solve the problem, we will follow the method covered by Pereira and Pinto [3]. Given that our problem is stochastic, we will first solve the deterministic case. Below is a general first stage problem,

$$\begin{aligned}
& \text{Min} && \mathbf{c}_1\mathbf{x}_1 + \alpha_1(\mathbf{x}_1) \\
& \text{subject to: } && \mathbf{A}_1\mathbf{x}_1 \geq \mathbf{b}_1.
\end{aligned}$$

We call $\mathbf{c}_1\mathbf{x}_1$ the immediate cost of decision $\mathbf{x}_1$ and $\alpha_1(\mathbf{x}_1)$ represents the future cost of decision $\mathbf{x}_1$. Below is how we define the future cost [3]:

$$\begin{aligned}
\alpha_1(\mathbf{x}_1) &= \text{Min} && \mathbf{c}_2\mathbf{x}_2 \\
& \text{subject to: } && \mathbf{A}_2\mathbf{x}_2 \geq \mathbf{b}_2 - \mathbf{E}_1\mathbf{x}_1.
\end{aligned} \tag{5}$$

One possible way to find the future cost of $\mathbf{x}_1$ is to discretise x_1 into trial values $\{\hat{\mathbf{x}}_{1i}, i = 1, \dots, n\}$. Then solving problem (5) for each trial value. However, this is expensive and falls to the curse of dimensionality. Instead, we will take the dual of problem (5),

$$\begin{aligned} \alpha_1(\mathbf{x}_1) &= \text{Max} \quad \pi(\mathbf{b}_2 - \mathbf{E}_1 \mathbf{x}_1) \\ &\text{subject to: } \pi \mathbf{A}_2 \leq \mathbf{c}_2. \end{aligned}$$

Where π is the vector of dual variables [3]. From linear programming theory, we know that the optimal solution to the above problem is equivalent to problem (5). Notice how the decision variable $\mathbf{x}_1$ is now in the objective function rather than the constraints, this means we can identify the constraint set $\pi \mathbf{A}_2 \leq \mathbf{c}_2$ before knowing decision $\mathbf{x}_1$ [3].

2.3 Solving the stochastic problem

The methodology will be the same as the deterministic problem where we assume an initial decision, then solve the following m second stage subproblems [3]. In terms of modelling reservoir systems, this looks like assuming we make some decision about reservoir (1) then solving the subproblem m -times for reservoir (2) (this means that reservoir (2) has m discrete levels of control). This gives us the following problem [3],

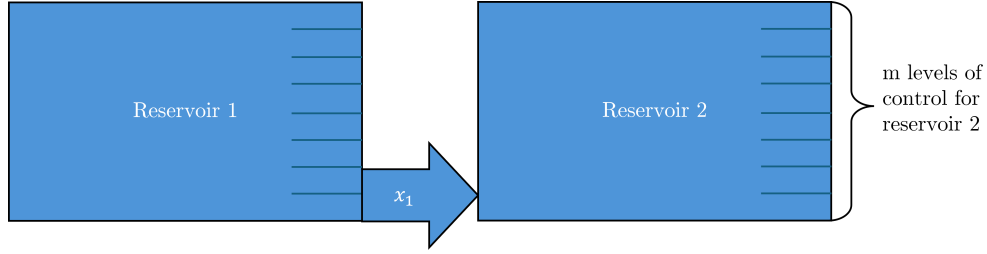


Figure 3: Visualisation of the 2 stage problem.

$$\begin{aligned} \text{Min} \quad & \mathbf{c}_1 \mathbf{x}_1 + p_1 \mathbf{c}_2 \mathbf{x}_{21} + p_2 \mathbf{c}_2 \mathbf{x}_{22} + \cdots + p_m \mathbf{c}_2 \mathbf{x}_{2m} \\ \text{subject to: } & \mathbf{A}_1 \mathbf{x}_1 \geq \mathbf{b}_1, \\ & \mathbf{E}_1 \mathbf{x}_1 + \mathbf{A}_2 \mathbf{x}_{21} \geq \mathbf{b}_{21}, \\ & \mathbf{E}_1 \mathbf{x}_1 + \mathbf{A}_2 \mathbf{x}_{22} \geq \mathbf{b}_{22}, \\ & \quad \quad \quad \vdots \\ & \mathbf{E}_1 \mathbf{x}_1 + \mathbf{A}_2 \mathbf{x}_{2m} \geq \mathbf{b}_{2m}. \end{aligned}$$

Again, $\mathbf{c}_1 \mathbf{x}_1$ represents the initial cost as a result of decision $\mathbf{x}_1$. Notice we have to account for all possible second stage actions, where p_i is the probability of subproblem i occurring. Similar to the deterministic case we will form the second stage subproblems [3],

$$\begin{aligned} \alpha_{1j}(\mathbf{x}_1) &= \text{Min} \quad \mathbf{c}_2 \mathbf{x}_{2j} \\ &\text{subject to: } \mathbf{A}_2 \mathbf{x}_{2j} \leq \mathbf{b}_{2j} - \mathbf{E}_1 \mathbf{x}_1. \\ &\quad \quad \quad \forall j = 1, \dots, m \end{aligned}$$

Our objective is to minimise the initial costs plus the expected value of the second-stage costs ($\sum_{j=1}^m p_j \mathbf{c}_2 \mathbf{x}_{2j}$) [3]. This leads us the expected future cost function $\hat{\alpha}_1(x_1)$ where we have taken the trial decision x_1 ,

$$\hat{\alpha}_1(\mathbf{x}_1) = \sum_{j=1}^m p_j \alpha_{1j}(\mathbf{x}_1)$$

Using this we can rewrite the first-stage problem in terms of the expected future cost function to get the following,

$$\begin{aligned} \text{Min} \quad & \mathbf{c}_1 \mathbf{x}_1 + \hat{\alpha}_1(\mathbf{x}_1) \\ \text{subject to: } & \mathbf{A}_1 \mathbf{x}_1 \geq \mathbf{b}_1 \end{aligned}$$

As shown in shown in the deterministic case we can solve the dual of the above problem to avoid expensive computational costs. Now we can define the backwards dynamic programming algorithm to solve the complete problem. To implement the following algorithm we will assume that the vectors $\{b_t, t = 1, \dots, T\}$ can be discretised into m scenarios $\{b_{tj}, j = 1, \dots, m\}$ with corresponding probabilities $\{p_j, j = 1, \dots, m\}$ [3]. Heuristically, the algorithm finds a lower and upper bound for expected running costs and will terminate when they are sufficiently close. The lower bound is found by finding the optimal value of the above problem, which can be solved by finding the dual. Whilst the upper bound is calculated by solving problem (6).

Algorithm 1 Multistage Stochastic Dual Dynamic Programming [3]

Step (a).

Define a set of trial decisions $\{\hat{\mathbf{x}}_{ti} \text{ for } i = 1, \dots, n, t = 1, \dots, T\}$.

Step (b)

for $t = T, T - 1, \dots, 2$ (backwards recursion) **do**

for for each trial decision $\hat{\mathbf{x}}_{ti}, i = 1, \dots, n$ **do**

for for each scenario $\mathbf{b}_{tj}, j = 1, \dots, m$ **do**

solve the optimisation problem for $t, \hat{\mathbf{x}}_{t-1i}$, and $\mathbf{b}_{tj}$

 Min $\mathbf{c}_t \mathbf{x}_t + \hat{\alpha}_t(\mathbf{x}_t)$

 subject to: $\mathbf{A}_t \mathbf{x}_t \geq \mathbf{b}_{tj} - \mathbf{E}_{t-1} \hat{\mathbf{x}}_{t-1i}$.

 Let π_{t-1ij} be the cost for scenario b_{t-1ij}

end for

end for

end for

Go to Step (a).

Using the Lagrange multiplier π_{t-1ij} we can calculate the expected cost of a given scenario b_{ji} , where p_{ij} is the associated probability using [3],

$$\hat{\pi}_{t-1i} = \sum_{j=1}^m p_{tj} \pi_{t-1ij}.$$

To ensure that the backwards dynamic programming method results in accurate optimal results, it is highly suggested that we should run forward simulations [4]. This involves selecting an initial decision $\mathbf{x}_1$, from which the one stage problem can be solved for all stages $(t, \dots, T)$. This can be done using the expected future costs function $\hat{\alpha}_t$.

2.4 Model Enhancements

As briefly mentioned before, one could alter the model to optimise other aspects of a hydro electrical system. There is a range of potential impacts a hydroelectric plant can have, we will show how to implement a range of new variables to help mitigate or reduce any problems that might occur. ICE is a government owned institution in Costa Rica, they developed a hydro-ecological model (RANA-ICA) this takes into account hydrological, social and biological demands [5]. This was crucial to ICE since they were

going to develop a reservoir system that could impact locals as they relied on a stable population of fish and high quality water in order to meet food requirements. The development of the new dam (gen-dam) was going to drastically restrict the movement of the fish and in turn the future population. They could partially address these problems by creating a dam on a similar nearby river, that would be maintained as a biodiverse aquatic habitat (we will call this eco-dam)

This only solves part of the problem, since some fish would still have to go through gen-dam. ICE created the following constraint, know as No Net Loss (NNL) [5]. To maintain water quality:

$$\mathbf{q}^1 * \mathbf{y} * \mathbf{p} + \sum (\mathbf{q}^2 - \mathbf{q}^o) * \mathbf{z} > \mathbf{q}^3 \mathbf{x},$$

where:

p : probability of the eco-dam being built without ICE's intervention, in order to meet the NNL conditions,

y : km of habitat that would be disconnected by the eco-dam,

x : km of habitat that would be disconnected by the gen-dam,

z : km of habitat across the area that flows into the eco-dam,

q : water quality:

q^1 : quality now above the eco-dam (water that is held back),

q^o : quality of the downstream before opening the eco-dam,

q^2 : quality of the downstream after opening the eco-dam,

q^3 : quality now before flooding between gen-dam and an existing dam.

The above was used to ensure the water quality would be maintained even after introducing the two new dams.

To maintain biodiversity [5]:

$$(\mathbf{a}^1 - \mathbf{a}^o) > \mathbf{a}^2,$$

where:

a^o : abundance in the upper lengths of the river leading to eco-dam before releasing water,

a^1 : abundance in the upper lengths of the river leading to eco-dam after releasing water,

a^2 : abundance between gen-dam and an existing dam.

This was specific to the ecological problem that ICE was facing. If one was adapting this model for Scottish systems, we might want to change where the data is recorded as we would be concerned with the abundance of salmon and trout.

3 Conclusion

We have seen how SDDP can be applied to the context of optimising a hydroelectric system. We started by working to minimise the running costs of a system, and the theory behind why this method is so successful. Then to show how one could implement SDDP into their system we covered the backwards algorithm using the theory we learned. To ensure that the large scale use of hydroelectric plants is sustainable we showed how we could create environmental constraints for our system. We will now take into account what we have covered to assess the suitability to create a hydroelectric system in Scotland.

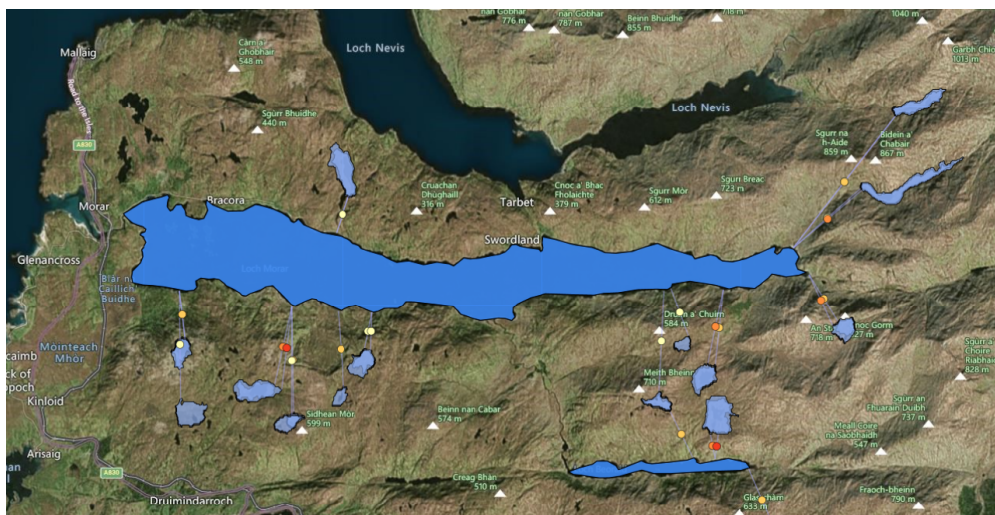


Figure 4: Potential Bluefield reservoirs around Loch Morar

Loch Morar is on the West coast of Scotland, with a length of 18.8km and reaching a depth of 310m. Loch Morar is the deepest body of freshwater in Britain [6]. Using figure 4 we can see there is plenty of potential to implement hydroelectric power stations in and around Loch Morar, but how suitable is it? Loch Morar is home to a wide array of aquatic life including salmon and trout as well as aquatic flora, which provide habitat for a range of smaller species [6]. Clearly if we were to implement hydroelectric stations here, we would need to be delicate with how we operate our system in order to preserve the biodiversity. To the South East of Loch Morar there is Loch Beoraid. There is potential for two independent systems that run from the peaks between Morar and Beoraid. Below we can see the capture range of Morar and Beoraid, both cover a significant portion of the surrounding area which would be ideal for creating new systems.

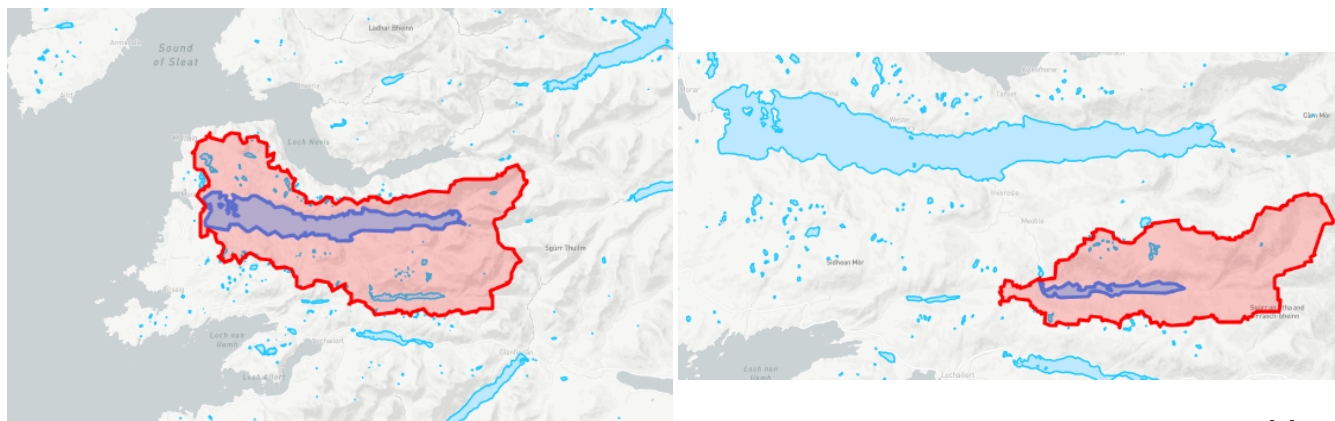


Figure 5: Catchment area of Loch Morar [7].

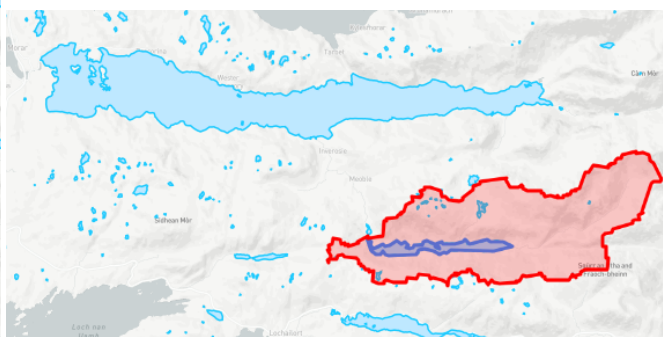


Figure 6: Catchment area of Loch Beoraid [7].

From our Bluefield sites between Morar and Beoraid we can see that the water source originates from the peaks and flows directly into on of the Lochs. Instead, if we connected the smaller reservoirs before leading into each Loch we would create a scenario suitable for SDDP. From this brief heuristic analysis, we can see there would be potential for new hydroelectric systems in Scotland.

References

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